

Skills Summary

Modeling Loci and Intersections with Conics

Use this reference while completing your worksheet or when studying.

Skill 1 — A circle from a distance condition

Locus definition. “Every point at distance r from (h, k) ” is the circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Expanding gives $x^2 + y^2 + Dx + Ey + F = 0$; completing the square takes you back. Square distances before you simplify — radicals almost never need to appear.

Example: A point stays 3 units from $(2, -5)$. Write its equation in expanded form.

Step 1. The condition is a fixed distance from a fixed point, which is the definition of a circle, so start from the squared distance formula rather than plotting points.

Step 2. $(x - 2)^2 + (y + 5)^2 = 9$ expands to $x^2 - 4x + 4 + y^2 + 10y + 25 = 9$.

$$\Rightarrow x^2 + y^2 - 4x + 10y + 20 = 0$$

Try it: A point stays 6 units from $(-1, 4)$. Write its equation in expanded form.

$$0 = 61 - 6x - 2y + x^2 + y^2 \quad \text{check:}$$

Skill 2 — A parabola from focus and directrix

Locus definition. A parabola is every point equally far from the focus and from the directrix.

With the vertex at the origin: $x^2 = 4py$ opens up or down with focus $(0, p)$ and directrix $y = -p$; $y^2 = 4px$ opens sideways with focus $(p, 0)$ and directrix $x = -p$. Here p is the vertex-to-focus distance.

Example: A parabola has vertex $(0, 0)$ and focus $(0, 2)$. Find $4p$.

Step 1. The focus is directly above the vertex, so the axis is vertical and p is just the vertical gap from vertex to focus.

Step 2. $p = 2$, so $4p = 8$ and the equation is $x^2 = 8y$, with directrix $y = -2$.

$$\Rightarrow 4p = 8$$

Try it: A parabola has vertex $(0, 0)$ and focus $(0, 7)$. Find $4p$.

$$28 = 4p \quad \text{check:}$$

(!) Watch out: the directrix sits on the *opposite* side of the vertex from the focus, the same distance away. Putting it on the same side gives a parabola opening the wrong way.

Skill 3 — Where two conics meet

Method. Substitute one equation into the other to get a single quadratic, solve it, then feed *every* root back to recover the partner coordinate.

The number of solutions of that quadratic is the number of crossings: two roots means two points, a repeated root means a tangent, no real root means the curves miss.

Example: Find where $y = x^2 + 1$ meets $y = 3x - 1$.

Step 1. Both equations already give y , so setting the right-hand sides equal removes y in one step.

Step 2. $x^2 + 1 = 3x - 1$, so $x^2 - 3x + 2 = 0$ and $(x - 1)(x - 2) = 0$; then $y = 3x - 1$ gives 2 and 5.

$\Rightarrow$ (1, 2) and (2, 5)

Try it: Find where $y = x^2 - 2$ meets $y = x$.

(1, -1) and (2, 2) check